

# Mykola Zhuk

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## 📁 Experience

### Software Analyst

Jun 2024 – Aug 2024 | Toronto

Accenture

- **Developed** AI proof-of-concept tools in **Python** using **OpenAI APIs** and **Jupyter** to support client Q&A and semantic search for retail brands, **improving customer-facing search by 20%**.
- **Built and enriched data pipelines** that ingested 50K+ product records, boosting search relevance and downstream model readiness, cutting **prototype time by 30%**.
- **Participated in iterative Agile sprints, code reviews**, and stakeholder demos to ship backlog features in close partnership with **cross-functional teams**.

## 🛠 Projects

### Backend Engineer

Oct 2025 – Mar 2026 | Remote

Tacmedbat (Volunteering)

- Built and maintained a **NestJS + MongoDB backend** with a localized **UA/EN public API**, aggregating **8 content collections** into **1 landing endpoint** that supports **10+ frontend content blocks**.
- Delivered **15+ backend features and fixes** across **Cloudinary** image handling (**7+ image use cases**), **RBAC, CORS, validation, email, archive/sort logic, and payment/contact integrations**.
- **Website Link** 📄

### Full-Stack Engineer 📄

Oct 2024 – Dec 2024 | Toronto

Scriptorium-CSC309 Project

- Built a secure, **Docker-based sandbox** supporting **real-time code execution** with enforced time & memory limits in **10+ languages** and deployed a full-stack app (React / Next.js / Prisma / PostgreSQL) on **AWS**.
- Designed **20+ REST API endpoints** with JWT-authenticated multi-role access for **40+ community code templates**. **Manually tested every endpoint** using **Postman**.
- Delivered a platform with **OAuth 2.0 (Google & GitHub)**, **Monaco editor** integration, and fork/search features; awarded 94% final grade.
- **Live Demo (AWS)** 📄

### AI/ML Engineer

Jun 2026 – Present | Remote, Toronto

Financial News Event Extraction Pipeline (Startup MVP Project)

- **Built a hybrid NLP/LLM extraction pipeline** across **20 corporate-event categories**, processing **2,936 articles** and producing **1,332 records, 1,328 schema-valid (99.7%)**.
- **Designed controlled experiments comparing candidate-selection strategies on a 300-article holdout**, improving precision 45.0% → 58.6% at ~95% recall and downstream lenient/strict F1 83.2% → 85.7% / 67.9% → 69.5%.
- **Cut inference usage** with cached classification, precision guards, and selective LLM invocation, reducing **model calls by 17.6%** and **token usage by 17.9%**.
- **Improved rare-event coverage** with classifier recovery and deterministic rules, raising candidate recall **75.7% → 87.1%** and reducing missed events **16 → 8** on a **200-article rare-event-heavy benchmark**.
- **Built automated validation and reliability checks** for the extraction pipeline, including schema validation, retry handling, and regression tests.

### Backend Engineer 📄

Oct 2024 – Dec 2024 | Toronto

ProjectOllon (Industry-Partnered MVP) - CSC301 Project

- Designed a relational schema with **Prisma ORM** to model 100+ files and store 2,000+ code components, **cutting full-scan time by 80%**.
- **Integrated OpenAI + Cohere LLMs** with policy-based filtering to auto-flag code complexity, ensuring secure & responsible suggestions, **cutting manual code review by 60%**.
- Built static-analysis checks catching **30% more style and duplication issues** pre-LLM, improving report precision and overall code quality for the partner team.

## Skills

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**Languages:** Python, Java, JavaScript/TypeScript, SQL, Bash

**Frameworks:** JUnit, pytest, Postman/REST API testing, React/Next.js, Node/Express, NestJS

**Cloud & DevOps:** Linux, Git, Docker, CI/CD (GitHub Actions/Jenkins), AWS (EC2), Jira (issues/triage), log collection & parsing basics

**Core Skills:** Debugging & log analysis, crash/stack-trace reading, reproduction environment setup (Docker/local), diagnostic scripts (Python/Bash), SQL diagnostics (joins/CTEs/window), Networking basics (HTTP/TCP/IP, DNS, TLS), Agile ceremonies, documentation & stakeholder updates

**Testing & Reliability:** unit testing (pytest/Jest), integration testing (API/DB), load-test basics, code reviews

## Education

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**Honours Bachelor of Science, Mathematics & CS**

Sep 2020 – Jun 2025 | Toronto

*University of Toronto*

Relevant Courses: Software Design, Machine Learning, Data Structures and Algorithms, Web Programming, System Programming, Computer Organization